ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-D 1.5]

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Intake Shutter Valve Operation Inspection

1. Connect the M-MDS to the DLC-2.
2. Switch the ignition ON (engine off).
3. Access and monitor the PID/DATA monitor item ISV_ACT and simulation item ISV_DSD. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
4. Verify that the ISV_ACT PID value increases linearly, while increasing the opening angle of the intake shutter valve using the active command mode function of ISV_DSD.
   - If the ISV_ACT PID does not increase according to the intake shutter valve opening angle, inspect using the following steps.
     1. Remove the intake shutter valve with connector connected. (See INTAKE SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
     2. Verify that the intake shutter valve opens and closes while it is operated using the active command mode function of the ISV_DSD.
        — If not verified, retest after cleaning the intake shutter valve and inspecting related wiring harness and connectors.
     3. If the intake shutter valve does not operate, replace the intake shutter valve. (See INTAKE SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - If the ISV_ACT PID increases increase linearly according to the intake shutter valve opening angle, the intake shutter valve is normal.

Brake Override System Operation Inspection

Note
- If the brake override system operates normally after performing the following inspection, the PCM detects DTC P2299:00.

1. Start the engine and run it is idling.
2. Verify that the engine speed becomes **approx. 1,000 rpm** under the following conditions.
   - Accelerator pedal position: **5 % or more**
   - Neutral (MTX)
   - N position (ATX)
   - Engine speed of **875 rpm or more**
   - Brake pedal depressed
     — If the engine speed becomes **approx. 1,000 rpm**, clear the PCM DTC using the M-MDS. (System operation is normal.) (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)
     — If the engine speed does not become **approx. 1,000 rpm**, inspect for the following parts, then repair or replace the malfunctioning part:
       - APP sensor (See ACCELERATOR PEDAL POSITION (APP) SENSOR INSPECTION [SKYACTIV-D 1.5].)
       - Neutral switch (MTX) (See NEUTRAL SWITCH INSPECTION [SKYACTIV-D 1.5].)
       - CPP switch (MTX) (See CLUTCH PEDAL POSITION (CPP) SWITCH INSPECTION [SKYACTIV-D 1.5].)
       - Brake switch (No.1 signal) (See BRAKE SWITCH INSPECTION.)
       - Brake switch (No.2 signal) (See BRAKE SWITCH INSPECTION.)

HP-EGR Control Valve Operation Inspection

1. Connect the M-MDS to the DLC-2.
2. Switch the ignition ON (engine off).
3. Access the PID/DATA monitor item HPEGR_ACT and simulation item HPEGR_DSD. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
4. Verify that the HPEGR_ACT PID value increases linearly, when the simulation item HPEGR_DSD increases from **0 %** to **approx. 90 %**.
   - If the HPEGR_ACT PID does not increase according to the HPEGR_DSD values, inspect using the following steps.
     1. Remove the HP-EGR control valve with connector connected. (See HP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
     2. Verify that the HP-EGR control valve opens and closes while it is operated using the active command mode function of the HPEGR_DSD.
        — If not verified, retest after inspecting HP-EGR control valve related wiring harness and connectors.
     3. If the HP-EGR control valve does not operate, replace the HP-EGR control valve. (See HP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - If the HPEGR_ACT PID increases increase linearly according to the HP-EGR control valve opening angle, the HP-EGR control valve is normal.

Exhaust Shutter Valve Operation Inspection

1. Connect the M-MDS to the DLC-2.
2. Switch the ignition ON (engine off).
3. Access and monitor the PID/DATA monitor item EXHSHUT_ACT and simulation item EXHSHUT_DSD. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
4. Verify that the EXHSHUT_ACT PID value increases linearly, while increasing the opening angle of the exhaust shutter valve using the active command mode function of EXHSHUT_DSD.
   - If the EXHSHUT_ACT PID does not increase according to the exhaust shutter valve opening angle, inspect using the following steps.
     1. Remove the exhaust shutter valve with connector connected. (See EXHAUST SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
     2. Verify that the exhaust shutter valve opens and closes while it is operated using the active command mode function of the EXHSHUT_DSD.
        — If not verified, retest after cleaning the exhaust shutter valve and inspecting related wiring harness and connectors.
        — If the exhaust shutter valve is damaged or stuck, replace the valve. (See EXHAUST SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
     3. If the exhaust shutter valve does not operate, replace the exhaust shutter valve actuator. (See EXHAUST SHUTTER VALVE ACTUATOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - If the EXHSHUT_ACT PID increases increase linearly according to the exhaust shutter valve opening angle, the exhaust shutter valve is normal.

LP-EGR Control Valve Operation Inspection

1. Connect the M-MDS to the DLC-2.
2. Switch the ignition ON (engine off).
3. Access the PID/DATA monitor item LPEGR_ACT and simulation item LPEGR_DSD. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
4. Verify that the LPEGR_ACT PID value increases linearly, when the simulation item LPEGR_DSD increases from **0 %** to **approx. 90 %**.
   - If the LPEGR_ACT PID does not increase according to the LPEGR_DSD values, inspect using the following steps.
     1. Remove the LP-EGR control valve with connector connected. (See LP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
     2. Verify that the LP-EGR control valve opens and closes while it is operated using the active command mode function of the LPEGR_DSD.
        — If not verified, retest after inspecting LP-EGR control valve related wiring harness and connectors.
     3. If the LP-EGR control valve does not operate, replace the LP-EGR control valve. (See LP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - If the LPEGR_ACT PID increases increase linearly according to the LP-EGR control valve opening angle, the LP-EGR control valve is normal.

Fuel Injector Operation Inspection

1. Connect the M-MDS to the DLC-2.
2. Start the engine and warm it up to normal operating temperature.
3. Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
4. Verify that DTCs P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00 and/or P1379:00 are not shown using the KOER self test.
   - If DTC P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00 and/or P1379:00 are shown, perform the DTC troubleshooting procedure. (See DTC TABLE [SKYACTIV-D 1.5].)
5. Verify that the engine speed drops or stalls when each cylinder fuel injector turns to off from on using the M-MDS active command mode function. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
   - If it cannot be verified, inspect the following for the suspected cylinder;
     — Fuel injector (See FUEL INJECTOR INSPECTION [SKYACTIV-D 1.5].)
     — Wiring harness for PCM—Fuel injector

Boost Control Operation Inspection

1. Connect the M-MDS to the DLC-2.
2. Start the engine and run it at idle.
3. Access and monitor PID/DATA monitor item VGTURB_ACT and simulation item VGTURB_DSD. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
4. Verify that the VGTURB_ACT PID value increases linearly, while increasing the active command mode function of VGTURB_DSD.

- If the VGTURB_ACT PID value does not increase linearly according to the active command function of the VGTURB_DSD value, inspect the followings:
  - — Turbocharger solenoid valve (See TURBOCHARGER SOLENOID VALVE INSPECTION [SKYACTIV-D 1.5].)
  - — Turbocharger actuator (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY INSPECTION [SKYACTIV-D 1.5].)
- If the VGTURB_ACT PID increases increase linearly according to the active command function of VGTURB_DSD value, boost control operation is normal.

A/C Cut-off Control System Inspection

1. Start the engine.
2. Turn the A/C switch and the fan switch on.
3. Verify that the A/C compressor magnetic clutch actuates.
   - If it does not actuate, go to symptom troubleshooting “NO.28 A/C DOES NOT WORK SUFFICIENTLY”. (See NO.28 A/C DOES NOT WORK SUFFICIENTLY [SKYACTIV-D 1.5].)
4. Fully open the throttle valve and verify that the A/C compressor magnetic clutch does not actuate for **2—5 s**.
   - If it actuates, inspect as follows:
     - — A/C relay (See RELAY INSPECTION.)
     - — Open or short to ground circuit in wiring harness and connectors (IG1 relay terminal C—A/C relay—PCM terminal 2AQ)
     - — A/C related parts
     - — APP1, APP2 PIDs (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See PCM INSPECTION [SKYACTIV-D 1.5].)

Engine Oil Solenoid Valve Operation Inspection

1. Verify that the oil pressure is normal. (See OIL PRESSURE INSPECTION [SKYACTIV-D 1.5].)
2. Connect the M-MDS to the DLC-2.
3. Start the engine.
4. Access the PID/DATA monitor items OIL_P_DUTY, RPM and ECT. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
5. When the following conditions are met, verify the oil pressure gauge value changes:
   - Engine coolant temperature (PID: ECT): **98 °C {208 °F} or more**
   - Engine speed: **3,000 rpm or more**
   - OIL_P_DUTY PID value is changing.
     - — If the oil pressure gauge value does not change:
       - Inspect the engine oil solenoid valve. (See ENGINE OIL SOLENOID VALVE INSPECTION [SKYACTIV-D 1.5].)
     - — If the oil pressure gauge value changes:
       - Engine oil solenoid valve is normal.